Functional Annotation of *cysI* (PP_2371, UniProt Q88KB9) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

cysI (locus **PP_2371**; UniProt **Q88KB9**) encodes the **sulfite reductase hemoprotein (SiRHP), the β subunit of assimilatory NADPH-dependent sulfite reductase (EC 1.8.1.2)**. Its primary catalytic function is the **six-electron reduction of sulfite (SO_3^{2-}) to sulfide (S^{2-}/H_2S)** at a unique **siroheme–[4Fe-4S] coupled cofactor**. This is the **terminal reductive step of the assimilatory sulfate-reduction pathway**, generating the reduced sulfur (sulfide) that is subsequently incorporated into L-cysteine (and thence methionine, coenzymes, and other sulfur metabolites). The enzyme is a **soluble cytoplasmic** protein that acts as the catalytic partner of the CysJ diflavin flavoprotein, which delivers electrons from NADPH. The functional assignment is supported by conserved domain architecture and sequence homology, and by decades of experimental biochemistry, crystallography, and genetics in the closely related enterobacterial and *Pseudomonas*/actinobacterial systems.

Identity verification: The gene symbol *cysI*, the UniProt description ("Sulphite reductase hemoprotein, beta subunit"), the InterPro/Pfam domains (PF01077 nitrite/sulfite reductase 4Fe-4S; PF03460 ferredoxin-like half-domain), and the cofactor keywords (4Fe-4S, Heme, Iron, Iron-sulfur, Oxidoreductase) are **fully mutually consistent** and match the canonical CysI hemoprotein of assimilatory sulfite reductase. No ambiguity was found. Where direct *P. putida* KT2440 biochemical studies of PP_2371 are lacking, function is inferred from strong orthology to experimentally characterized CysI proteins.

2. Molecular Identity and Domain Architecture

Property	Value (UniProt Q88KB9)
Gene / locus	*cysI* / PP_2371
Length / mass	550 aa / 62.2 kDa
Organism	*Pseudomonas putida* KT2440 (taxid 160488)
Keywords	4Fe-4S, Heme, Iron, Iron-sulfur, Metal-binding, Oxidoreductase
GO (function)	GO:0016491 oxidoreductase; GO:0020037 heme binding; GO:0051539 [4Fe-4S] binding; GO:0046872 metal-ion binding
Domains	2× "Nitrite/Sulfite reductase 4Fe-4S" (Pfam PF01077) + 2× "ferredoxin-like" (Pfam PF03460)
Cross-refs	KEGG ppu:PP_2371; BioCyc PPUT160488:G1G01-2534-MONOMER; AlphaFoldDB Q88KB9

The protein has the characteristic **internally duplicated two-domain repeat** of the nitrite/sulfite-reductase (NIR/SIR) superfamily, and its C-terminus contains the canonical cysteine-rich cofactor-ligating motif (…G**C**MNA**C**GHHH…**C**…) that supplies the four cysteine ligands to the [4Fe-4S] cluster, one of which is the **bridging thiolate shared with the siroheme iron**. There is no signal peptide or transmembrane segment, consistent with a soluble cytoplasmic localization.

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3. Primary Function: Catalysis and Substrate Specificity

The CysI hemoprotein is the **catalytic subunit** that binds substrate and carries out its reduction. The reaction is:

> **SO_3^{2-} + 6 e^- + 6–8 H^+ → S^{2-} (H_2S) + 3 H_2O**

Key features:

- **Six-electron chemistry with no released intermediates.** The active site of the *E. coli* hemoprotein "is exquisitely designed to catalyze the six-electron reductions of sulfite to sulfide and nitrite to ammonia" (Crane, Siegel & Getzoff, 1997, P 9315848). The enzyme therefore has a **secondary nitrite-reductase activity** (nitrite → ammonia), reflecting shared active-site chemistry across the NIR/SIR family, but its **physiological substrate is sulfite** in the cysteine-biosynthetic (assimilatory) context.
- **Push–pull mechanism.** "The substrate evolves through a push-pull mechanism, where electron transfer is coupled to three dehydration steps" (Askenasy & Stroupe, 2020, P 32851831). An extensive network of positively charged residues, ordered waters, and siroheme carboxylates binds, polarizes, and protonates the anionic substrate at the siroheme sixth axial position (Crane et al., 1997, P 9315848).
- **Assimilatory holoenzyme.** In *E. coli*, "the flavoprotein and hemoprotein components of … NADPH-sulfite reductase are encoded by *cysJ* and *cysI*, respectively" (Wu, Siegel & Kredich, 1991, P 1987123). Electrons flow **NADPH → FAD → FMN (CysJ diflavin flavoprotein) → siroheme/[4Fe-4S] (CysI) → sulfite**. Recent cryo-EM confirms the two-subunit organization: "SiR has two subunits: an NADPH, FMN, and FAD-binding diflavin flavoprotein and a siroheme/Fe[4S4]-containing hemoprotein" (Ghazi Esfahani et al., 2025, P 40140349). The classical holoenzyme stoichiometry is $\alpha_8\beta_4$ ($CysJ_8CysI_4$).

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4. Cofactors: The Siroheme–[4Fe-4S] Coupled Center

The defining feature of CysI is its bimetallic active site:

- **Siroheme covalently coupled to a [4Fe-4S] cluster via a shared cysteine.** "The hallmark of sulfite reductase is its catalytic center made of an iron-containing porphyrinoid called siroheme that is covalently coupled to a [4Fe-4S] cluster through a shared cysteine ligand" (Askenasy & Stroupe, 2020, P 32851831). Crystallography across oxidation states shows "a bridging cysteine thiolate supplied by the protein always covalently links the siroheme (iron isobacteriochlorin) to the Fe4S4 cluster, facilitating their ability to transfer electrons to substrate" (Crane, Siegel & Getzoff, 1997, P 9315848). Sulfite binds directly to the siroheme iron; the [4Fe-4S] cluster tunes the redox potential and relays electrons.
- **Cofactor dependency on siroheme biosynthesis (CysG).** Siroheme is made from uroporphyrinogen III by methylation, oxidation, and iron insertion. "*cysG* mutants cannot reduce sulfite to sulfide and require a source of sulfide or cysteine for growth" (Kolko et al., 2001, P 11114933), and *cysG* "encodes a uroporphyrinogen III methyltransferase required for the synthesis of siroheme, a cofactor for the hemoprotein" (Wu et al., 1991, P 1987123). Without siroheme, CysI accumulates as inactive apo-protein. (*P. putida* KT2440 encodes the siroheme synthase machinery required to activate CysI.)

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5. Biological Process and Pathway Context

CysI catalyzes the **last dedicated reductive step of assimilatory sulfate reduction**, the pathway supplying reduced sulfur for biosynthesis:

> sulfate (SO_4^{2-}) → **APS** (ATP sulfurylase, CysN/CysD) → **PAPS/sulfite** (APS/PAPS reductase, CysH; in *Pseudomonas* an APS reductase route) → **sulfide** (**sulfite reductase, CysJI**) → **L-cysteine** (O-acetylserine sulfhydrylase / cysteine synthase, CysK/CysM)

Genetic evidence directly ties *cysI* to sulfide production: - In *Zymomonas mobilis*, inactivation of the sulfite-reductase genes *cysIJ* rendered mutants "unable to produce detectable H_2S," and sulfite could not restore it (Tan et al., 2013, P 23086550). - In *Corynebacterium glutamicum*, mutational analysis showed *cysI* (among the *cys* cluster) "were demonstrated to be involved in the reduction of inorganic sulphur compounds" (Rückert et al., 2005, P 16159395).

The sulfide produced is condensed with O-acetyl-L-serine to form L-cysteine, the entry point for all sulfur-containing biomolecules (cysteine, methionine, glutathione, Fe-S clusters, thiamine, biotin, coenzyme A). In enteric bacteria the *cys* regulon (including *cysJIH*) is controlled by the LysR-type activator **CysB** with N-acetylserine as inducer and sulfide/cysteine as anti-inducers (Borum & Monty, 1976, P 1107320); an analogous sulfur-responsive regulatory logic operates in *Pseudomonas*.

5.1 KT2440-specific genomic and pathway organization (this study)

Database mapping of the *P. putida* KT2440 genome (KEGG) provides direct, organism-specific confirmation and refines the pathway picture:

- **Exact assignment:** KEGG annotates **ppu:PP_2371** = **K00381, "sulfite reductase (NADPH) hemoprotein beta-component" [EC 1.8.1.2]**, in pathway **ppu00920 (Sulfur metabolism)** and modules **M00176 (Assimilatory sulfate reduction, sulfate ⇒ H_2S)** and **M00616 (Sulfate–sulfur assimilation)**. Chromosomal position: complement(2,707,556–2,709,208), 1,653 bp (550 aa).
- **Not an operon with *cysJ*:** The genes immediately flanking PP_2371 (PP_2370, unknown; PP_2372, putative sugar-binding protein) are unrelated. Thus, unlike the *E. coli cysJIH* operon, KT2440 *cysI* is **genomically dispersed** from its flavoprotein partner. The CysJ

diflavin flavoprotein (KO K00380) maps to **PP_0860 / PP_1703** elsewhere on the chromosome.

- **PAPS-independent (APS-reductase) route:** KT2440 encodes an **APS reductase (CysH, K00390 = PP_2328)** and ATP-sulfurylase component **CysD (K00957 = PP_1303)**, but **no APS kinase (cysC, K00860) and no PAPS reductase** were found. This indicates that *P. putida* reduces activated sulfate as **APS directly to sulfite (bypassing PAPS)** — consistent with the [4Fe-4S] APS reductase characterized in the related *P. aeruginosa* (Kim et al., 2004, P 15491155). CysI (PP_2371) then performs the final **sulfite → sulfide** step, and the sulfide is delivered to **cysteine synthase / O-acetylserine sulfhydrylase (CysK, K01738 = PP_4571)** to form L-cysteine.

Pathway in KT2440: SO_4^{2-} → (ATP sulfurylase, CysDN) → **APS** → (**APS reductase CysH/PP_2328**) → **sulfite** → (**sulfite reductase CysI/PP_2371** + **CysJ flavoprotein PP_0860/PP_1703**) → **sulfide** → (**CysK/PP_4571** + **O-acetylserine**) → **L-cysteine**.

Note on pathway specialization: Assimilatory sulfite reductase (CysI/SiRHP) is mechanistically and evolutionarily distinct from **dissimilatory** sulfite reductase (DsrAB), which uses sulfite as a terminal respiratory electron acceptor (Colman et al., 2022, P 35122336); both, however, share the siroheme–[4Fe-4S] catalytic strategy (Askenasy & Stroupe, 2020, P 32851831). PP_2371 belongs to the **assimilatory** class, consistent with *P. putida*'s aerobic biosynthetic sulfur metabolism.

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6. Subcellular Localization

CysI is a **soluble cytoplasmic** enzyme. It carries no signal peptide or transmembrane helix (UniProt Q88KB9), and the assimilatory NADPH-sulfite reductase holoenzyme is a soluble cytoplasmic complex composed of the CysJ flavoprotein and CysI hemoprotein (Ghazi Esfahani et al., 2025, P 40140349; Wu et al., 1991,

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P 1987123

It therefore performs its reaction in the cytoplasm, drawing reducing equivalents from cytoplasmic NADPH — in contrast to membrane-bound/periplasmic respiratory nitrite/sulfite reductases (e.g., cytochrome *c* nitrite reductase NrfA).

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7. Supported and Refuted Hypotheses

Supported - H1: PP_2371/*cysI* encodes sulfite reductase hemoprotein (β subunit). **Supported** (domain architecture, cofactor keywords, orthology; PMIDs 9315848, 1987123, 40140349). - H2: The enzyme catalyzes the six-electron reduction of sulfite to sulfide. **Supported** (PMIDs 9315848, 40140349, 32851831). - H3: Catalysis occurs at a siroheme–[4Fe-4S] coupled center via a bridging cysteine, using a push-pull mechanism. **Supported** (PMIDs 32851831, 9315848). - H4: The gene product functions in the assimilatory sulfate-reduction/cysteine-biosynthesis pathway; loss abolishes H_2S. **Supported** (PMIDs 23086550, 16159395). - H5: The product is a soluble cytoplasmic hemoprotein partnering the CysJ diflavin flavoprotein. **Supported** (PMIDs 40140349, 1987123; UniProt features).

Refuted / excluded - That PP_2371 is a **dissimilatory** (respiratory) sulfite reductase — excluded by family/domain type and aerobic assimilatory context. - That the physiological substrate is nitrite — nitrite reduction is a documented *in vitro* side activity but the biological role is sulfite reduction for cysteine biosynthesis.

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8. Limitations and Future Directions

- **No direct biochemical/structural study of PP_2371 itself** was found; the functional assignment rests on strong orthology to experimentally characterized CysI/SiRHP proteins (primarily *E. coli*) and on genetic evidence from other bacteria. Direct enzymatic assay, cofactor quantitation, and structure determination of the *P. putida* protein would confirm the inference.
- **Operon structure and regulation in KT2440** (e.g., *cysJIH* organization, CysB-type control, sulfur-starvation induction) were not experimentally verified here and warrant genomic/transcriptomic confirmation.

- **Electron-donor partner in *Pseudomonas*.** While *E. coli* uses the NADPH diflavin flavoprotein CysJ, some organisms use ferredoxin/flavodoxin donors; the exact physiological reductant for PP_2371 should be confirmed.

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9. Key References

- Crane BR, Siegel LM, Getzoff ED. Structures of the siroheme- and Fe4S4-containing active center of sulfite reductase… *Biochemistry* 1997. PMID **9315848**.
- Askenasy I, Stroupe ME. The Siroheme-[4Fe-4S] Coupled Center. 2020. PMID **32851831**.
- Ghazi Esfahani et al. Structure of dimerized assimilatory NADPH-dependent sulfite reductase… 2025. PMID **40140349** (also P 38915618).
- Wu JY, Siegel LM, Kredich NM. High-level expression of *E. coli* NADPH-sulfite reductase (cysJ/cysI; cysG/siroheme). 1991. PMID **1987123**.
- Tan et al. H_2S formation by *cysIJ*-inactivated *Zymomonas mobilis*. 2013. PMID **23086550**.
- Rückert et al. *C. glutamicum* assimilatory sulphate-reduction *cys* cluster. 2005. PMID **16159395**.
- Kolko et al. Alternative siroheme synthesis; *cysG* required for sulfite reduction. 2001. PMID **11114933**.
- Colman et al. Structural evolution of dissimilatory sulfite reductase (contrast). 2022. PMID **35122336**.
- Borum & Monty. CysB regulation of cysteine biosynthetic enzymes. 1976. PMID **1107320**.
